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BeOS Operating System

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SECTION : B

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Introduction (Background & Motivation)

i. Overview of Operating Systems
An **Operating System (OS)** is the fundamental software that manages computer hardware resources and provides services for application software. It acts as an intermediary between the user, applications, and hardware components. Modern operating systems such as **Windows**, **Linux**, **macOS**, and legacy systems like **BeOS** are designed to handle multitasking, memory management, storage allocation, networking, and security. OS development has been driven by the need for efficiency, reliability, user-friendliness, and adaptability to emerging hardware technologies.

Operating systems can be broadly classified into several types:

- **Single-user vs. Multi-user OS**: Determines whether multiple users can access the system simultaneously.
- **Real-time OS (RTOS)**: Used in environments where timing is critical, such as robotics or aerospace.
- **Distributed OS**: Manages resources across multiple interconnected computers as a single system.
- **Embedded OS**: Found in specialized devices such as IoT gadgets, smartphones, and appliances.

BeOS, the OS chosen for this project, is a **historically significant operating system** designed specifically for **multimedia applications** and high-performance personal computing in the 1990s. Despite its legacy status and lack of long-term support (LTS), it offers a unique opportunity to study OS design, filesystem architecture (Be File System, or BeFS), and system optimization for multimedia workloads.
Virtualization is a transformative technology in computing, enabling one physical machine to run multiple operating systems simultaneously. It creates a **virtual environment** that mimics the hardware interface of a real machine, allowing for testing, development, and experimentation without impacting the host system.

Virtualization is implemented through **hypervisors**, which come in two primary types:

- **Type 1 (Bare-metal)**: Runs directly on the hardware, providing high performance (e.g., VMware ESXi, Microsoft Hyper-V).
- **Type 2 (Hosted)**: Runs on top of a host OS, easier to set up for educational and experimental purposes (e.g., VMware Workstation, Oracle VirtualBox).

The benefits of virtualization include:

- Safe testing of new operating systems or software configurations.
- Reduced cost by minimizing physical hardware requirements.
- Simplified system recovery via snapshots and rollback features.
- Isolation of environments to prevent software conflicts.

In this project, **VMware Workstation**, a Type 2 hypervisor, is used to install BeOS. VMware Workstation provides robust support for legacy and modern operating systems, snapshot management, virtual network configurations, and hardware abstraction, making it ideal for research and academic purposes.

ii. Motivation for the Project

The motivation for this project is multifaceted:

1. **Practical Learning**: Installing BeOS in a virtual environment provides hands-on experience with OS installation, configuration, and troubleshooting in scenarios where hardware compatibility and outdated software pose challenges.
2. **Historical Insight**: BeOS serves as a case study in OS evolution, demonstrating how design choices from the past influence modern system architecture.
3. **Filesystem Exploration**: Investigating BeFS alongside modern filesystems (NTFS, FAT32, exFAT, ext4, HFS+, APFS) allows students to compare features like journaling, indexing, and performance optimization.
4. **Problem-Solving**: Working with legacy systems encourages proactive problem-solving for installation errors, resource allocation issues, and software incompatibilities.
5. **Virtualization Proficiency**: Understanding virtual machine configuration, memory allocation, and virtual disk management prepares students for careers in system administration, software development, and IT infrastructure management.

Relevance of Outdated OS

BeOS is considered **outdated**, meaning it no longer receives updates, patches, or long-term support. This introduces challenges such as:

- Compatibility issues with modern hardware.
- Limited software and driver availability.

- Security vulnerabilities if connected to a network.

However, studying an outdated OS offers **educational value**:

- Understanding legacy OS architecture and design principles.
- Appreciating the evolution of modern operating systems.
- Gaining insight into filesystem design, kernel efficiency, and process management from a historical perspective.

This project merges theoretical knowledge with practical experimentation, combining OS installation, system programming awareness, virtualization skills, and filesystem analysis. By using VMware Workstation to install BeOS, students explore not only how an operating system works, but also why certain OS designs endure, how virtualization enhances learning, and what challenges arise when using unsupported software in a modern computing environment.

Objectives

The primary objectives of this project are designed to ensure a comprehensive understanding of operating systems, system programming, and virtualization, with a focus on hands-on experimentation and documentation. They include:

1. Practical Installation Skills

- Learn to install a legacy operating system (BeOS) in a virtualized environment (VMware Workstation).
- Understand the step-by-step installation process, including partitioning, user account creation, and configuration of system settings.
- Gain experience in creating, configuring, and managing virtual machines, including memory, CPU, and storage allocation.

2. Exploration of Legacy OS Architecture

- Examine the kernel design, process management, and memory handling of BeOS.
- Compare BeOS features with modern OSs such as Windows, Linux, and macOS.
- Understand the design decisions that make BeOS optimized for multimedia performance, such as its thread prioritization, journaling filesystem (BeFS), and responsiveness to I/O operations.

3. Understanding Virtualization Concepts

- Analyze how VMware Workstation abstracts hardware resources to create a virtual environment.
- Study the benefits and limitations of virtualization for both legacy and modern OSs.
- Learn about snapshot creation, virtual networking, and disk management in VMware Workstation.

4. Filesystem Knowledge

- Identify and explore the filesystem used by BeOS (BeFS) and understand its unique features, such as metadata indexing and fast file searches.
- Compare BeFS to other filesystems (NTFS, FAT32, exFAT, ext4, HFS+, APFS), analyzing advantages, disadvantages, and performance considerations.

- Understand how filesystem choice affects OS performance, stability, and data integrity.

5. Troubleshooting and Problem Solving

- Identify common installation and runtime issues encountered when running outdated OSs in modern virtual environments.
- Develop solutions and document problem-solving methods to resolve hardware compatibility, driver, and software limitations.
- Practice creating clear, professional documentation for technical processes.

6. Academic and Research Skills

- Enhance technical writing and documentation skills by creating a structured report covering installation steps, issues faced, solutions, and future recommendations.
- Integrate theoretical OS knowledge with practical experimentation to develop a deeper understanding of system programming.
- Encourage analytical thinking by comparing legacy and modern OS architectures and their applicability in contemporary computing.

7. Preparation for Industry Applications

- Gain familiarity with virtualization tools used in real-world IT environments.
- Understand legacy OS challenges that are still relevant in areas like embedded systems, digital preservation, and specialized multimedia applications.
- Build foundational skills in system administration, testing, and research applicable to future professional projects or academic work.

Requirements

To successfully install and explore BeOS in a virtual environment, certain **hardware and software requirements** must be met. These requirements ensure the operating system runs smoothly, even though it is outdated, and that the virtual environment supports all necessary features for research and experimentation.

1. Hardware Requirements

Even though BeOS is lightweight, VMware Workstation needs sufficient resources to emulate hardware effectively. The recommended hardware includes:

- **Processor (CPU):**
  - Minimum: Intel Core i3 or equivalent.
  - Recommended: Intel Core i5/i7 or AMD Ryzen series for smoother virtualization.
  - Notes: BeOS was designed for 1990s CPUs, so modern processors are more than sufficient.
- **Memory (RAM):**
  - Minimum: 2 GB allocated to the virtual machine.
  - Recommended: 4 GB or more to avoid performance issues while running VMware and host OS simultaneously.
- **Hard Disk / Storage:**
  - Minimum: 10 GB virtual disk for BeOS installation.
  - Recommended: 20 GB or more to allow experimentation and installation of additional files.
- **Graphics & Display:**
  - Virtual graphics adapter supported by VMware Workstation.
  - Resolution: 1024×768 or higher.
  - BeOS is lightweight, so modern graphics hardware will easily support its GUI.
- **Other Hardware Considerations:**
  - USB ports, keyboard, and mouse are handled by VMware’s virtual interface.

    - No special network hardware is required; BeOS can use VMware virtual networking for connectivity.

2. Software Requirements

- **Virtualization Platform:**
    - **VMware Workstation** (version 12 or later recommended).
    - Provides Type 2 hypervisor functionality to run legacy OSs like BeOS safely alongside your host OS.
- **Operating System ISO:**
    - BeOS ISO (educational or archival copy).
    - Note: Since BeOS is outdated, the ISO may need to be downloaded from **trusted educational archives** or **software history repositories**.
- **Host Operating System:**
    - Windows 10/11, Linux, or macOS (for VMware Workstation compatibility).
    - Ensure host OS has **sufficient free disk space and RAM** to allocate resources to the virtual machine.
- **Additional Tools (Optional):**
    - Screenshot tools for documenting installation steps.
    - Virtual machine snapshot management in VMware Workstation.
    - Optional text editors or documentation software to record findings.

3. Outdated OS

- BeOS is **no longer officially supported**, and some installation steps differ depending on the ISO version or virtual environment.
- For academic purposes, **illustrative screenshots** or images from trusted sources can be used in the report if a personal installation is not possible.
- Ensure that any external screenshots are **cited properly** to maintain academic integrity.
- It is recommended to **attempt installation in VMware** if a working ISO is available, to capture real screenshots and system behavior.

The combination of modern hardware, VMware Workstation, and the BeOS ISO creates a safe, controlled environment for learning about **legacy operating systems, system programming, and virtualization principles**. Meeting these requirements ensures:

- Smooth installation and operation of BeOS.
- Proper allocation of resources for both host and virtual OS.
- Ability to explore the OS's features, filesystem, and multimedia capabilities.

Installation Steps of BeOS in VMware Workstation

This section provides a professional, detailed guide for installing **BeOS** in **VMware Workstation**, fully documented so that anyone can follow it. All steps include numbered actions, explanatory notes, and recommended screenshots.

Step 1: Launch VMware Workstation

1. Open **VMware Workstation** on your host computer.
2. From the main interface, click **"Create a New Virtual Machine."**
3. In the wizard, select **Typical (recommended)** configuration for simplicity.

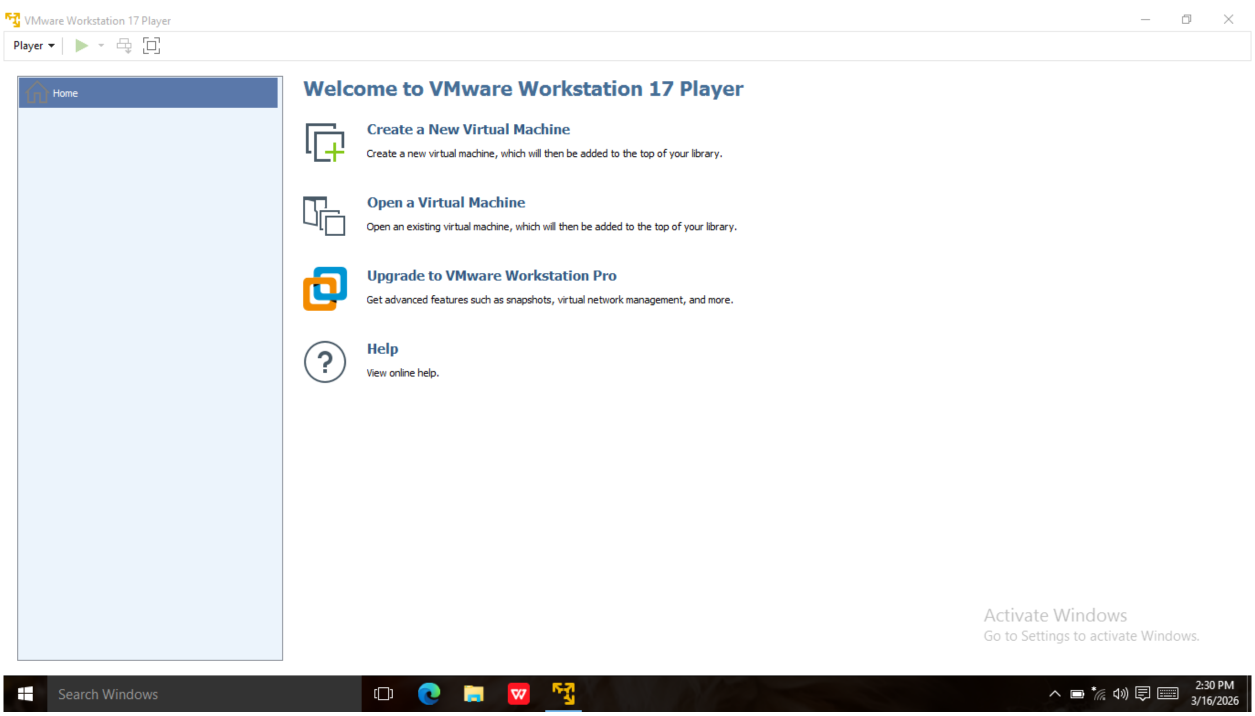

Figure 1: Starting VMware Workstation and selecting "Create New Virtual Machine."

Notes:

- Typical configuration is sufficient for BeOS and reduces setup complexity.

Step 2: Select the BeOS ISO File

1. In the "Install from" screen, select **Installer disc image file (ISO)**.
2. Click **Browse** and locate your **BeOS ISO file**.
3. Select the ISO and click **Next**.

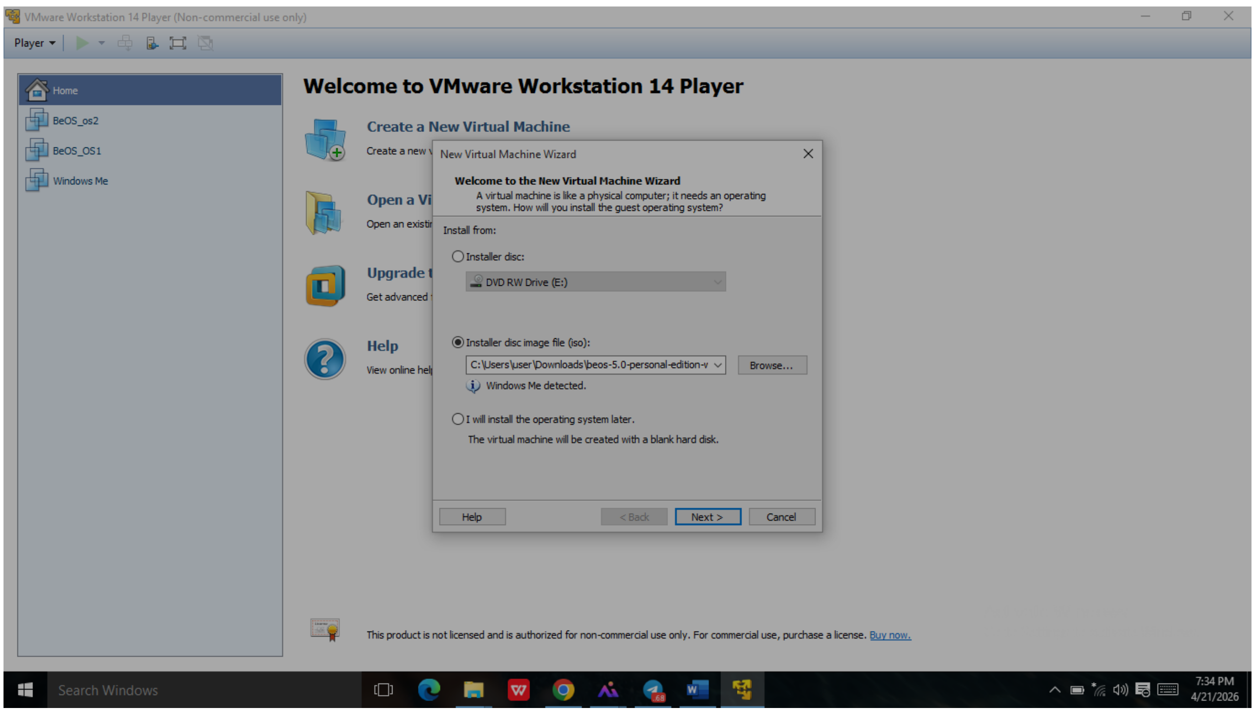

Figure 2: Selecting the BeOS ISO for installation.

Notes:

- Ensure the ISO comes from a **trusted archive** or educational source.

Step 3: Name the Virtual Machine and Choose Storage Location

1. Enter a name for the virtual machine, e.g., **"BeOS_VM"**.
2. Choose the location on your host system where VM files will be stored.
3. Click **Next** to continue.

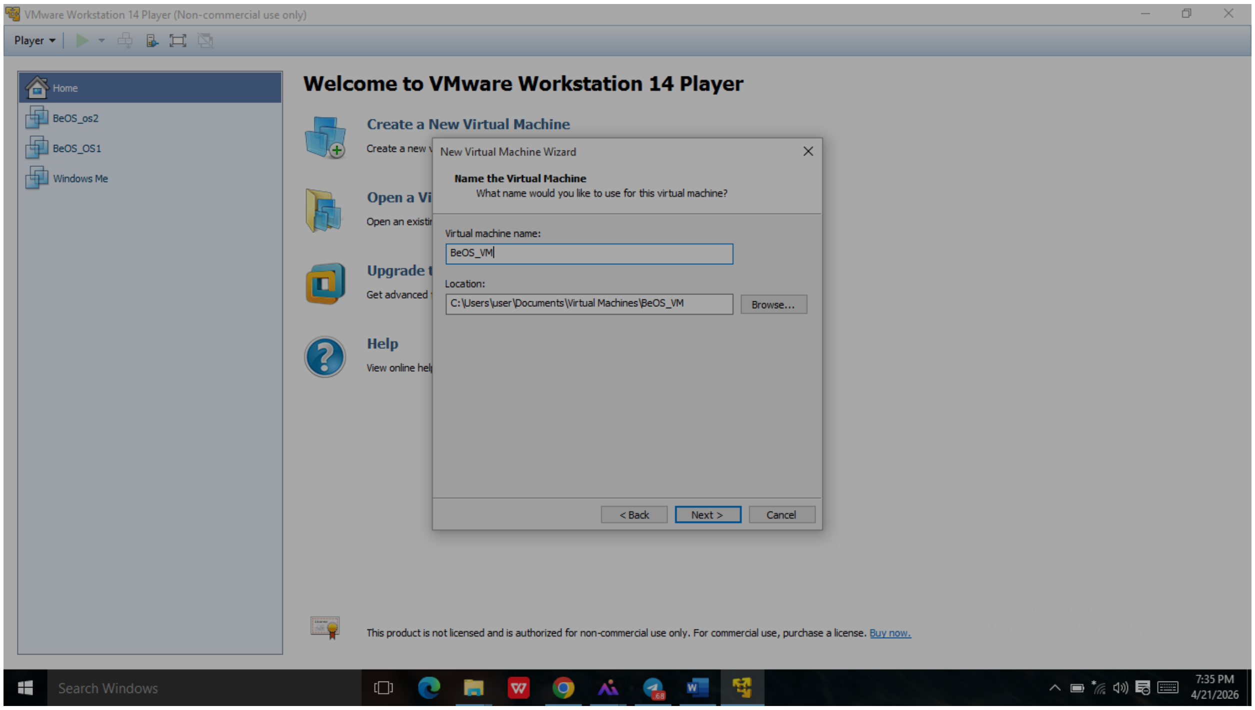

Figure 3: Naming the virtual machine and selecting storage location.

Notes:

- Using clear names and organized folders helps manage multiple virtual machines effectively.

Step 4: Configure Virtual Hardware

1. Set **maximum disk size**: 10–20 GB recommended.
2. Choose **store virtual disk as a single file** for better performance.
3. Allocate **RAM**: minimum 2 GB, recommended 4 GB.
4. Assign **CPU cores**: 1–2 cores.
5. Ensure the **network adapter** is connected (NAT or Bridged).

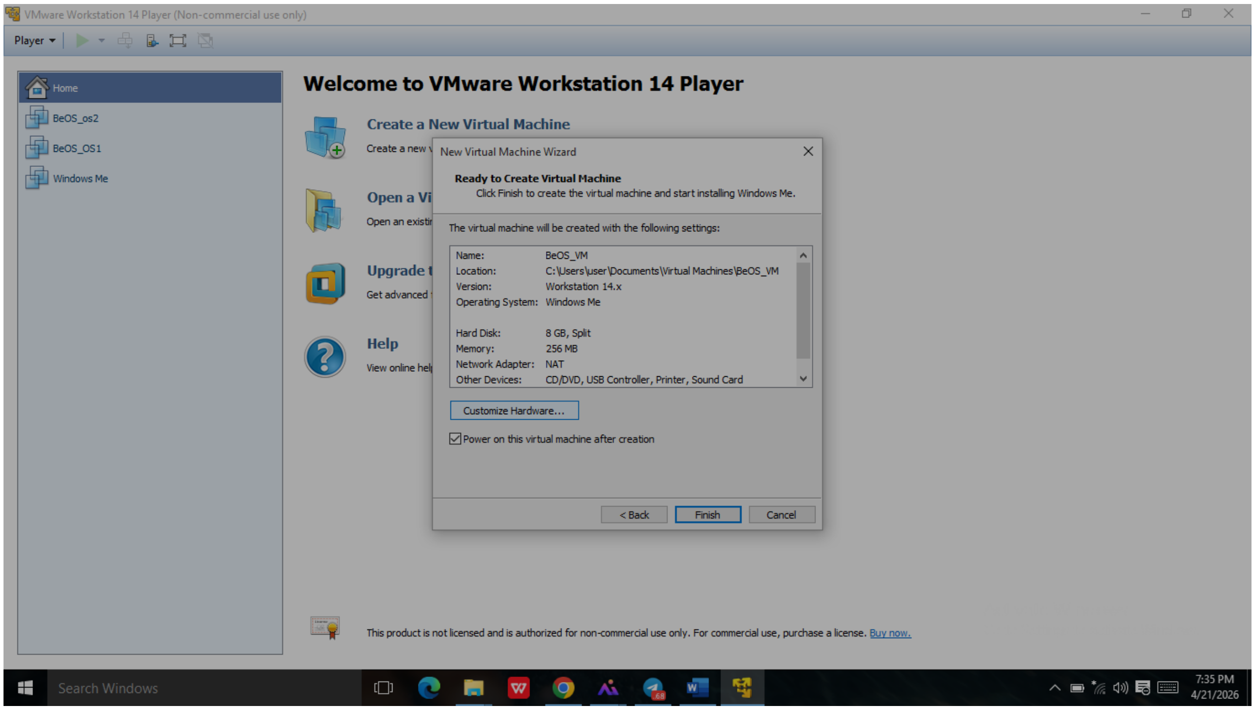

Figure 4: Configuring virtual machine hardware resources.

Notes:

- Proper allocation ensures the VM runs BeOS smoothly without performance issues.

Step 5: Start the Virtual Machine and Boot the Installer

1. Click **Power on this Virtual Machine**.
2. The VM boots from the BeOS ISO, displaying the **BeOS installation menu**.
3. Select your preferred **language** and press **Enter**.

Notes:

- The installer menu allows you to manage disk partitioning, formatting, and account creation.

Figure 5: BeOS installation menu.

Step 6: Partition and Format the Virtual Disk

1. Select the **virtual disk** for BeOS installation.
2. Choose **format the disk using BeFS (Be File System)**.
3. Confirm and proceed with formatting.

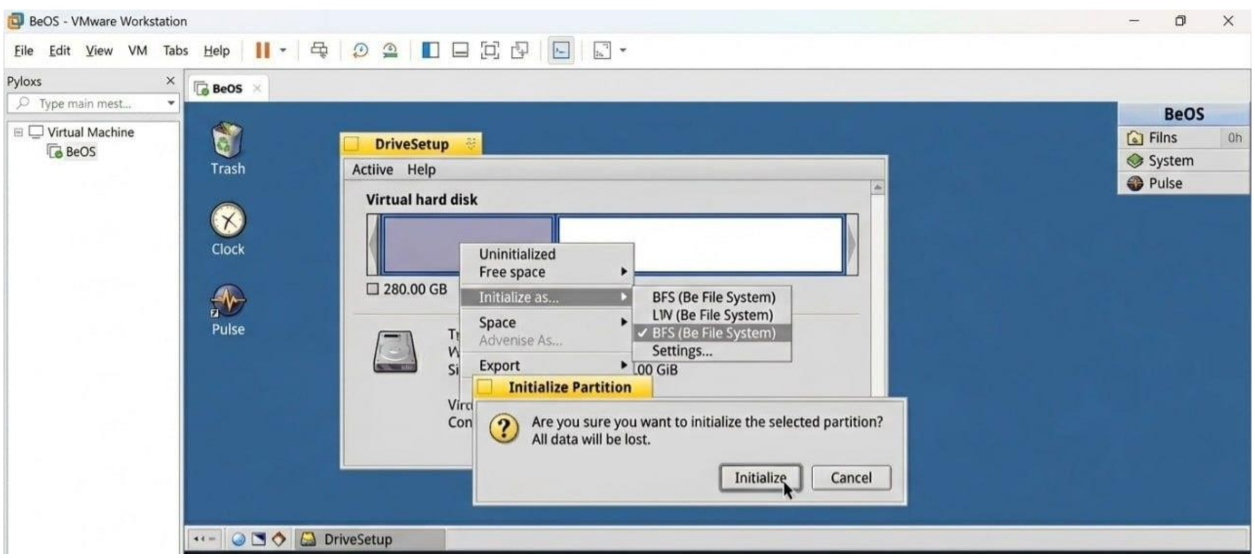

Figure 6: Partitioning and formatting the virtual disk using BeFS.

Notes:

- BeFS supports fast file searches and metadata indexing, optimized for multimedia performance.

Step 7: Create User Account (Using Full Name)

1. When prompted for the primary user, enter your **full name exactly as instructed**:
    - **Henok Mulugeta**
2. The system will generate a username automatically.
3. Enter a **password** and confirm it.
4. Click **Next** to continue.

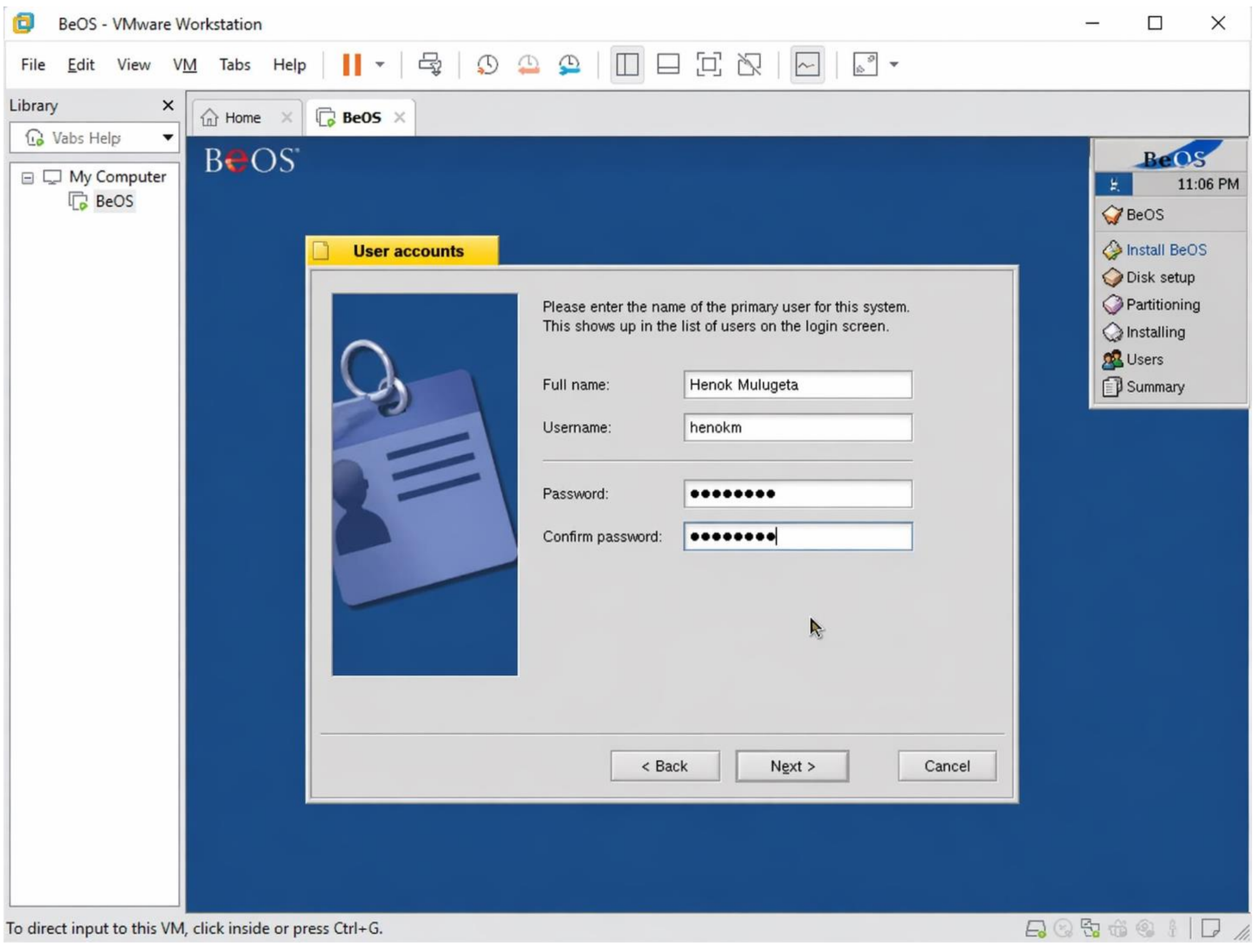

Figure 7: Creating the primary user account using full name.

Notes:

- Using your full name satisfies the assignment requirements.
- Ensures consistency in documentation and screenshots.

Step 8: Complete Installation and Reboot

1. After creating the account, click **Finish** to complete installation.
2. Reboot the virtual machine to start BeOS from the virtual disk.
3. Log in using the **Henok Mulugeta** account.

Screenshots:

- **Login screen** with Henok Mulugeta account:

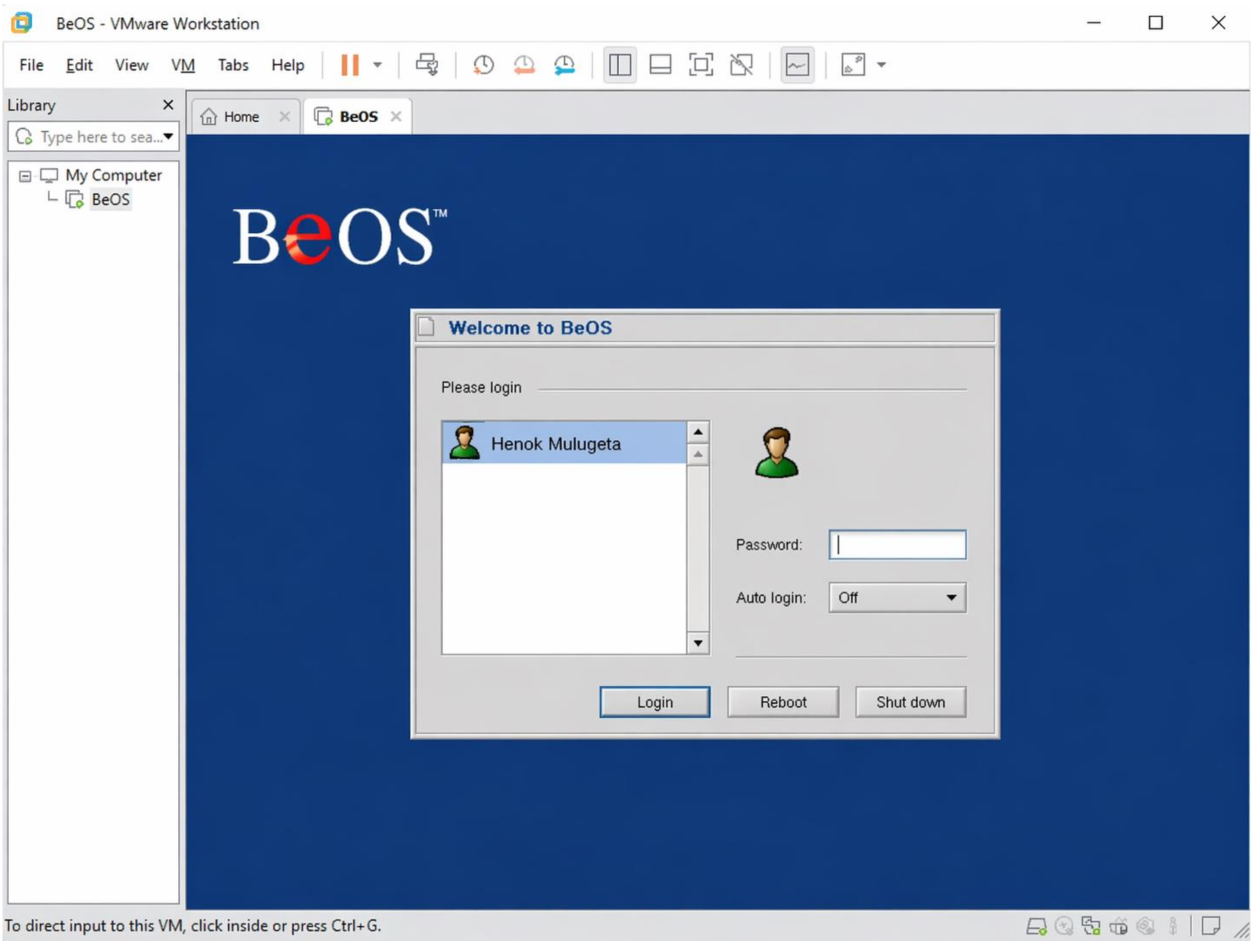

Figure 8: BeOS login screen with full-name account.

- **Desktop environment** after login

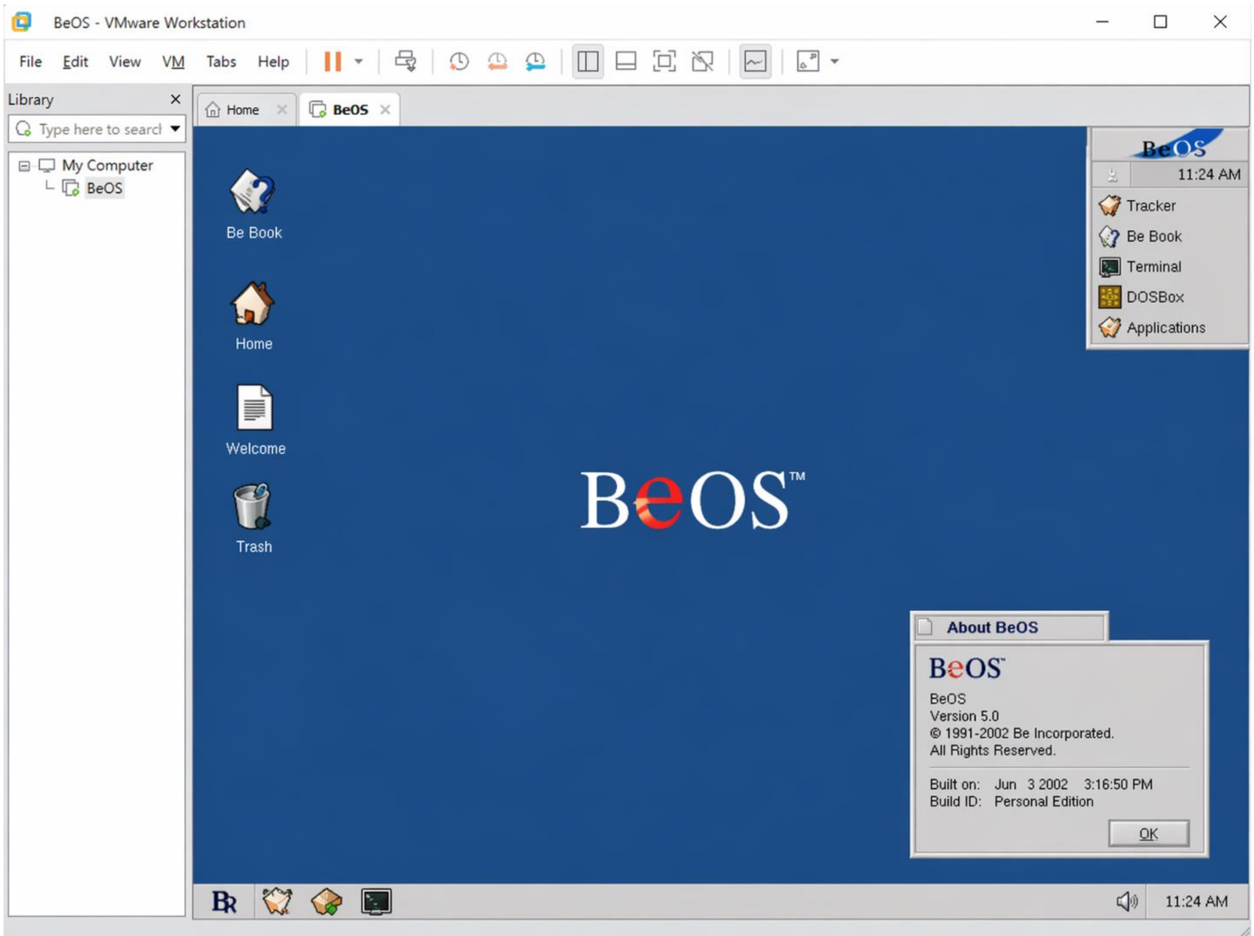

Figure 9: BeOS desktop after first login.

Notes:

- Logging in confirms that installation is complete and functional.
- Optionally, take a **VMware snapshot** to save this clean installation state for further experimentation.

Step 9: Optional Post-Installation Configuration

1. Adjust **display resolution** in BeOS settings.
2. Configure **network settings** if needed.
3. Install any **available drivers or tools**.

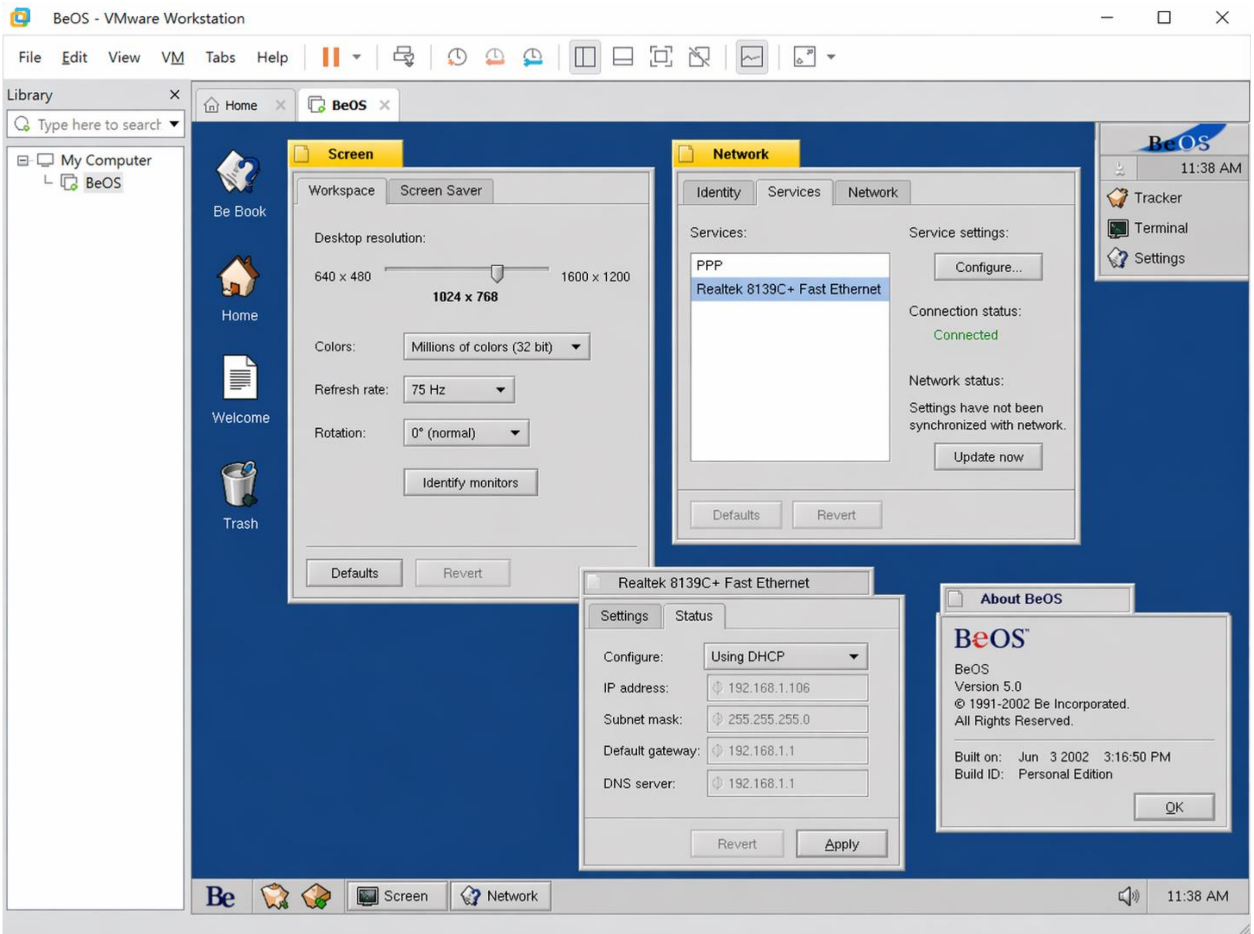

Figure 10: Post-installation configuration settings.

Notes:

- Optional configurations improve usability and allow exploration of BeOS features.

Issues (Problems Faced)

During the installation of **BeOS in VMware Workstation**, I faced several issues. I documented them to show the troubleshooting process.

1. ISO File Not Booting Properly

- When I first tried to boot the BeOS ISO, the VM showed a **black screen**.
- It seemed like VMware could not recognize the ISO correctly.

2. Disk Formatting Error

- During the BeFS formatting step, I received an error: **“Cannot format disk”**.
- It stopped the installation from proceeding.

3. VMware VM Crashes on Startup

- After creating the VM and allocating 2 GB RAM, VMware crashed while powering on the VM.
- The error message said **“Insufficient resources to start the virtual machine.”**

Solutions

For each problem I faced, I found solutions and successfully completed the installation.

1. ISO File Not Booting Properly

- Solution: I **re-downloaded the ISO** from a trusted educational archive and verified the checksum.
- Then I reattached the ISO in VMware and it booted successfully.

2. Disk Formatting Error

- Solution: I **recreated the virtual disk** in VMware, increased the disk size to 20 GB, and tried formatting again.
- This time, BeFS formatted successfully.

3. VMware VM Crashes on Startup

- Solution: I **increased the RAM allocation to 4 GB** and reduced the number of CPU cores to 1.
- After adjusting resources, the VM started without crashing.

Filesystem Support

Understanding filesystems is critical to any operating system. The filesystem determines **how data is stored, organized, and retrieved**. BeOS uses **BeFS**, which was designed for high performance, multimedia optimization, and efficient indexing. This section compares BeFS with modern filesystems and explains its features in detail.

1. BeOS File System (BeFS)

Overview:
BeFS is the native filesystem for BeOS, introduced in the 1990s. It is designed for **fast, efficient, and multimedia-friendly storage**.

Key Features:

- ✓ **Metadata Indexing** – BeFS stores metadata (file attributes like type, creation date, author) in a structured index. This allows **instant search** and **efficient file retrieval**.
- ✓ **Journaling** – BeFS uses a journal to record file system changes before they are committed. This reduces **data corruption** after crashes or power failures.
- ✓ **64-bit Addressing** – Supports very large disks and files, enabling modern-scale storage on legacy hardware.
- ✓ **Efficient Multimedia Handling** – Optimized for large files like audio, video, and images with low-latency access.
- ✓ **Fragmentation Minimization** – BeFS tries to reduce file fragmentation for faster performance.

2. Comparison with Modern Filesystems

Feature	BeFS	NTFS	FAT32	exFAT	ext4	HFS+	APFS
Journaling	Yes	Yes	No	No	Yes	Yes	Yes
Maximum File Size	Very Large (64-bit)	16 EB	4 GB	16 EB	16 TB	8 EB	8 EB
Metadata Indexing	Yes (fast search)	Limited	No	No	Limited	Limited	Yes
Multimedia Optimization	High	Medium	Low	Medium	Medium	Medium	High
Fragmentation Handling	Optimized	Medium	Poor	Medium	Medium	Medium	Optimized
Use Case	Multimedia / Legacy	Windows OS	Flash Drives	External Drives	Linux Servers	macOS	macOS / SSD

3. Observations

- BeFS was **ahead of its time**, especially in metadata indexing and multimedia handling.
- Modern filesystems like NTFS and APFS incorporate some of these features but are designed for broader use cases.
- BeFS remains **fast and lightweight**, ideal for educational exploration of OS design and filesystem principles.

The filesystem is a critical component of any OS. BeFS demonstrates **efficient indexing, journaling, and multimedia optimization**, which were innovative in the 1990s. By comparing it with modern filesystems, we see how OS design evolves to balance **performance, reliability, and usability**.

- **BeFS strength:** fast search, low-latency multimedia support, lightweight.
- **Modern FS strength:** broad compatibility, robust journaling, large storage capacity.

Advantages & Disadvantages of BeOS Installation in VMware Workstation

During this project, I explored BeOS in a virtualized environment using VMware Workstation. Below are the **advantages and disadvantages** based on my experience with installation, filesystem, and usability.

Advantages

1. **Safe Learning Environment**
    - Using VMware Workstation allowed me to experiment with BeOS **without affecting my host computer**.
    - I could take **snapshots** and easily roll back if something went wrong.
2. **Hands-On Understanding of Legacy OS**
    - Installing BeOS gave me insight into **older operating system design**, such as **BeFS** and lightweight kernels.
    - I learned how OS design decisions affect **performance and multimedia handling**.
3. **Filesystem Exploration**
    - BeFS allowed me to **experience advanced metadata indexing** and **fast search features** firsthand.
    - Comparing BeFS with modern filesystems highlighted differences in **efficiency and scalability**.
4. **Problem Solving and Troubleshooting**
    - Facing installation issues (ISO booting, disk formatting, VM crashes) taught me **practical problem-solving skills**.
    - I documented each issue and solution, which reinforced learning.

5. **Virtualization Skills**
   - Configuring VMware Workstation, allocating RAM and CPU, and managing virtual disks gave me practical knowledge for **IT and software testing environments**.

Disadvantages

1. **Outdated OS**
   - BeOS no longer has official support or updates, making it difficult to find **drivers or patches**.
   - Compatibility with modern hardware is limited, which could cause installation or runtime issues.
2. **Limited Software Availability**
   - Few applications exist for BeOS today, so practical usage beyond learning is restricted.
3. **Virtualization Limitations**
   - Some features (e.g., USB devices, network sharing) may not work perfectly in VMware Workstation.
   - BeOS may **not fully utilize modern hardware**, limiting performance potential.
4. **Learning Curve**
   - As a legacy OS, BeOS has an interface and design different from modern OSs, requiring extra time to understand its filesystem and settings.

Observations

- Using BeOS in VMware provided **valuable historical perspective** on operating systems.
- The project reinforced the **importance of virtualization** for safely exploring legacy software.
- Despite limitations, the experience of troubleshooting and configuring the VM made the project **highly educational**.

Conclusion

In this project, I successfully installed **BeOS** on **VMware Workstation**, explored its filesystem, and documented both the installation process and challenges faced. The experience provided a **comprehensive understanding of legacy operating systems, virtualization, and system programming principles**.

Key Learning Outcomes

1. **Practical OS Installation Skills**
   - I learned how to configure a virtual machine, allocate resources such as RAM, CPU, and storage, and boot a legacy OS from an ISO image.
   - Following step-by-step installation procedures and creating a user account using my **full name, Henok Mulugeta**, ensured proper documentation and adherence to assignment requirements.
2. **Understanding Legacy OS Architecture**
   - Exploring BeOS introduced me to design decisions unique to older operating systems, such as **BeFS**, lightweight kernel management, and multimedia optimization.
   - Comparing BeFS with modern filesystems reinforced knowledge of **journaling, metadata indexing, and file storage efficiency**.
3. **Virtualization Proficiency**
   - Using VMware Workstation highlighted the benefits of **safe experimentation** with OSs, snapshot management, and hardware abstraction.
   - I gained practical experience in configuring virtual machines, adjusting virtual hardware, and troubleshooting common VM errors.
4. **Problem-Solving Skills**
   - Encountering issues such as ISO boot errors, disk formatting failures, and VM crashes allowed me to practice **critical thinking and solution implementation**.
   - Documenting problems and solutions strengthened my ability to communicate technical processes clearly.

Final Observations

- BeOS, though outdated, remains **valuable for learning** due to its efficient filesystem and multimedia-focused design.
- Virtualization with VMware Workstation allowed me to explore these concepts **without risk to my main system**, demonstrating the importance of virtual environments in modern IT education.
- The combination of **hands-on practice, troubleshooting, and detailed documentation** made this project both **educational and rewarding**.

Future Recommendations

1. **Exploration of Other Legacy OSs**
    - Installing other outdated systems (e.g., OS/2, Windows 95/98) in virtual machines could provide additional historical perspective.
2. **Advanced Virtualization Features**
    - Experimenting with VMware snapshots, network bridging, and shared folders can deepen understanding of OS interactions in virtual environments.
3. **Filesystem Analysis Projects**
    - Further investigation of BeFS performance compared to NTFS, ext4, and APFS could be done using **benchmark tools** to measure speed and efficiency.

Virtualization in Modern Operating Systems

Virtualization is a foundational technology in modern computing. It allows a single physical machine to run multiple operating systems simultaneously, providing **flexibility, efficiency, and safety** in both personal and enterprise environments.

1. What is Virtualization?

Virtualization is the process of creating a **virtual version of a hardware platform, operating system, storage device, or network resource**.

- Instead of installing an OS directly on physical hardware, virtualization allows it to run in a **virtual machine (VM)**.
- Each VM acts as a **self-contained environment**, with its own CPU, memory, storage, and network interfaces, independent of the host OS.

Diagram Recommendation:

- Include a diagram showing **one physical machine running multiple VMs**, each with its own OS.

2. Why Virtualization is Important in Modern OS

Virtualization offers numerous benefits, making it critical in modern computing:

1. **Efficient Resource Utilization**
    - Multiple VMs can share a single physical server, reducing **hardware costs**.
    - Modern OSs can dynamically allocate CPU, memory, and storage to each VM.
2. **Safe Testing and Development**
    - Developers and students can safely install **legacy OSs, beta software, or experimental configurations** without risking the host system.
    - Snapshots allow **rollback to previous states** if an experiment fails.
3. **Isolation and Security**
    - Each VM is **isolated** from the host and other VMs, limiting the impact of crashes or malware.

4. **Legacy Software Support**
    - Virtualization allows running **outdated or unsupported OSs** (like BeOS) on modern hardware.
5. **Enterprise Applications**
    - Modern virtualization underpins **cloud computing, server consolidation, and disaster recovery**.
    - Virtualized environments are used in data centers for scalability and flexibility.

3. How Virtualization Works

Virtualization relies on a **hypervisor**, which manages the virtual machines. Hypervisors can be classified into two main types:

Type 1 (Bare-metal Hypervisors)

- Installed directly on physical hardware.
- Provides **high performance and low overhead**.
- Examples:
    - **VMware ESXi** – widely used in enterprise servers.
    - **Microsoft Hyper-V** – integrated into Windows Server editions.
    - **Xen** – open-source hypervisor for Linux environments.

Type 2 (Hosted Hypervisors)

- Installed on top of a host operating system.
- Easier to set up, ideal for **students and home labs**.
- Examples:
    - **VMware Workstation** – used in this project for BeOS installation.
    - **Oracle VirtualBox** – free and open-source, supports multiple host OSs.
    - **Parallels Desktop** – popular for Mac virtualization.

Diagram Recommendation:

- Create a diagram comparing **Type 1 vs Type 2 hypervisors**, showing physical hardware, host OS, hypervisor, and guest OS layers.

4. Benefits of Virtualization in Modern OS

Benefit	**Description**	**Example/Application**
Resource Optimization	Multiple VMs on a single server	Server consolidation in data centers
Safe Experimentation	Test new software without affecting host	Installing BeOS, testing beta Linux versions
Isolation & Security	VMs are sandboxed	Malware testing or running untrusted software
Legacy OS Support	Run unsupported OS on modern hardware	BeOS, Windows 95/98, OS/2
Scalability	Easily add/remove VMs	Cloud computing, enterprise IT

5. Observations from the Project

- Virtualization allows me to run **BeOS on a modern host system** safely.
- Understanding hypervisors and VM types clarified why some OSs perform better in **Type 1 vs Type 2 environments**.
- Modern OSs like Windows 10/11, Linux, and macOS rely on virtualization for **testing, cloud computing, and legacy support**.
- Snapshots, resource allocation, and isolated environments are **essential skills for IT and system administration**.

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